Software Project Management (SPM)

Software Project Management (SPM) is the process of **planning, organizing, leading, and controlling** a software project to achieve specific goals within **time, cost, and quality constraints**.

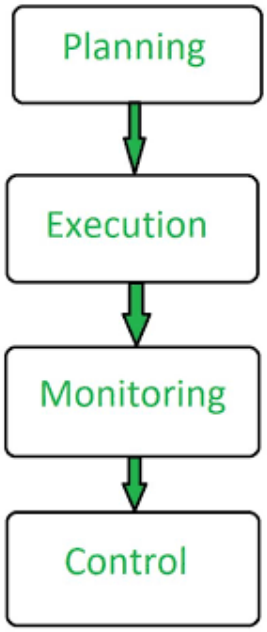
It ensures that:

- The project is delivered **on time**
- The **budget is maintained**
- The software meets **user requirements and quality standards**

Objectives of Software Project Management

- Deliver the **right product with required quality**
- Minimize **risks during development**
- Ensure efficient use of **resources (people, tools, time)**
- Handle **change management** effectively
- Achieve **customer satisfaction**

Process in SPM



<u>Key Activities in Software Project Management</u>

1. Project Planning

- Defines **scope, objectives, and deliverables**
- Estimates **time, cost, and resources**
- Selects suitable development models:
    - Agile
    - Waterfall
- Forms the foundation for the entire project.

2. Project Monitoring and Control

- Tracks **progress, quality, and cost**
- Compares actual performance with planned schedule
- Identifies **deviations and risks**
- Take **corrective actions** to keep the project on track.

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3. Team Management

- Assigns tasks and defines **roles and responsibilities**
- Ensures **effective communication and coordination**
- Motivates team members for better productivity
- Handles conflicts within the team.

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4. Risk Management

- Identifies potential risks such as:
    - Schedule delays
    - Budget overruns
    - Scope creep
- Analyzes risk impact and probability

- Develops **mitigation strategies**
- Continuously monitors risks throughout the project.

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5. Quality Management

- Defines **quality standards and guidelines**
- Conducts **reviews, testing, and audits**
- Ensures the final product meets **customer expectations**.

6. Customer and Stakeholder Management

- Collects and analyzes **requirements and feedback**
- Maintains regular communication with stakeholders
- Ensures alignment between **project output and customer needs**.

Challenges in Software Project Management

- Frequently changing requirements
- Errors in time and cost estimation
- Poor team coordination and communication
- Technological limitations
- Conflicts among stakeholders.

Tools Used in Software Project Management

Project Management Tools

- Microsoft Project
- Trello
- Jira

Communication Tools

- Slack
- Zoom

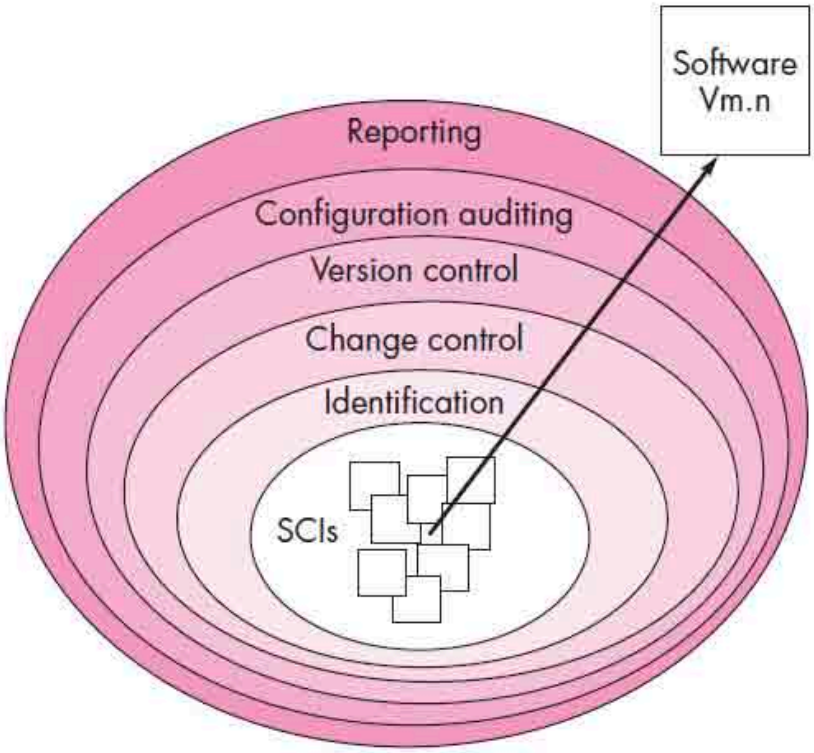
Time Tracking Tools

- Toggl
- Clockify

Software Configuration Management (SCM)

Software Configuration Management

Software Configuration Management (SCM) is the process of **managing, tracking, and controlling changes** in software during its development life cycle. It helps to make sure that **all changes in software are recorded, controlled, and organized properly**.



Need for SCM

SCM is important because software projects involve **many changes and multiple developers**.

Reasons:

- Avoid confusion between different versions
- Track changes made in code
- Manage team work efficiently
- Prevent errors due to uncontrolled changes
- Maintain consistency in the system

Objectives of SCM

- Control changes in software
- Maintain multiple versions of software
- Ensure product integrity and consistency
- Improve team coordination
- Provide traceability of changes

Key Activities of SCM

1. Configuration Identification

This step identifies all the important items in the software.

Example:

- Source code
- Design documents
- Test cases

These are called **Configuration Items (CI)**.

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2. Version Control

Version control manages different versions of software.

Example:

- Version 1.0
- Version 2.0
- Version 2.1

It helps developers:

- Track changes
- Go back to previous versions
- Work on the same project without conflict

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3. Change Control

Change control ensures that all changes are **properly reviewed and approved**.

Process:

1. Request for change
2. Analyze impact
3. Approve or reject
4. Implement change

Prevents unwanted or risky changes.

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4. Configuration Status Accounting

This activity records and tracks the status of all configuration items.

Example:

- Which version is current?
- What changes are made?
- Who made the change?

Helps in project tracking.

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5. Configuration Auditing

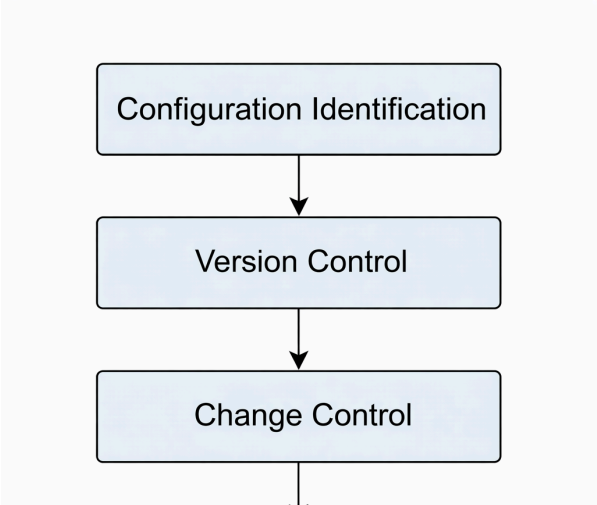
Auditing checks whether the software is correct and meets requirements.

Types:

- Functional Audit → checks functionality
- Physical Audit → checks documents and files

Ensures quality and correctness.

<u>SCM Process Flow</u>



SCM Tools

Some tools used in SCM:

- Git
- SVN (Subversion)
- Mercurial

These tools help in:

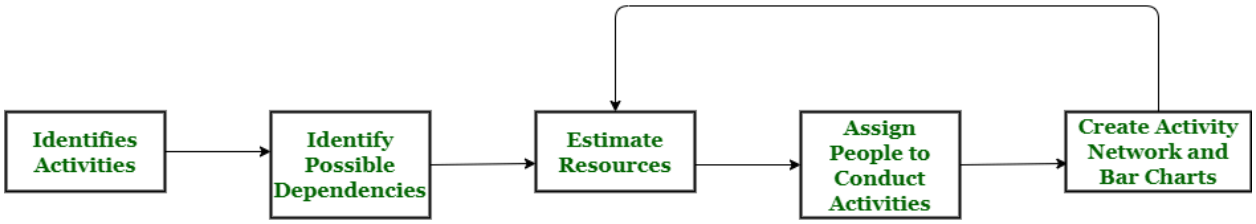
- version control
- collaboration
- tracking changes

Advantages of SCM

- Better control over software changes
- Easy team collaboration
- Reduced errors and conflicts
- Easy maintenance and updates
- Improved software quality

Project Scheduling

Project Scheduling is the process of **planning, organizing, and managing tasks, timelines, and resources** to ensure that a software project is completed **on time and within budget**.



Project Scheduling Process

Objectives / Goals of Project Scheduling

- Efficient allocation of resources
- Identification of task dependencies
- Determination of the critical path
- Monitoring progress and delays
- Ensuring the project stays on track

Key Elements of Project Scheduling

1. Work Breakdown Structure (WBS)

- Breaks the project into smaller, manageable components
- Helps in assigning responsibilities
- Makes tracking and control easier

2. Task Estimation

- Determines the effort and duration required for each task
- Techniques used:
    - Expert judgment
    - Analogy-based estimation
    - PERT (Program Evaluation and Review Technique)

3. Task Dependencies

- Defines the relationship between tasks
- Types:
    - Finish-to-Start (FS)
    - Start-to-Start (SS)
    - Finish-to-Finish (FF)

4. Milestones

- Important checkpoints in the project timeline
- Examples:
    - Completion of design
    - Prototype development
    - Final delivery

5. Critical Path Method (CPM)

- Identifies the **longest sequence of dependent tasks**
- Determines the minimum project completion time
- Delay in critical tasks leads to delay in the entire project

6. Gantt Charts

- Graphical representation of the project schedule

- Displays:
    - Task durations
    - Start and end dates
    - Dependencies
    - Progress tracking

Benefits of Project Scheduling

- Improves project visibility and control
- Ensures better resource utilization
- Enhances time management
- Helps in risk identification and proactive planning

Tools for Project Scheduling

- MS Project
- Trello
- Asana
- Jira
- Wrike
- Gantt Chart tools
- PERT Chart tools